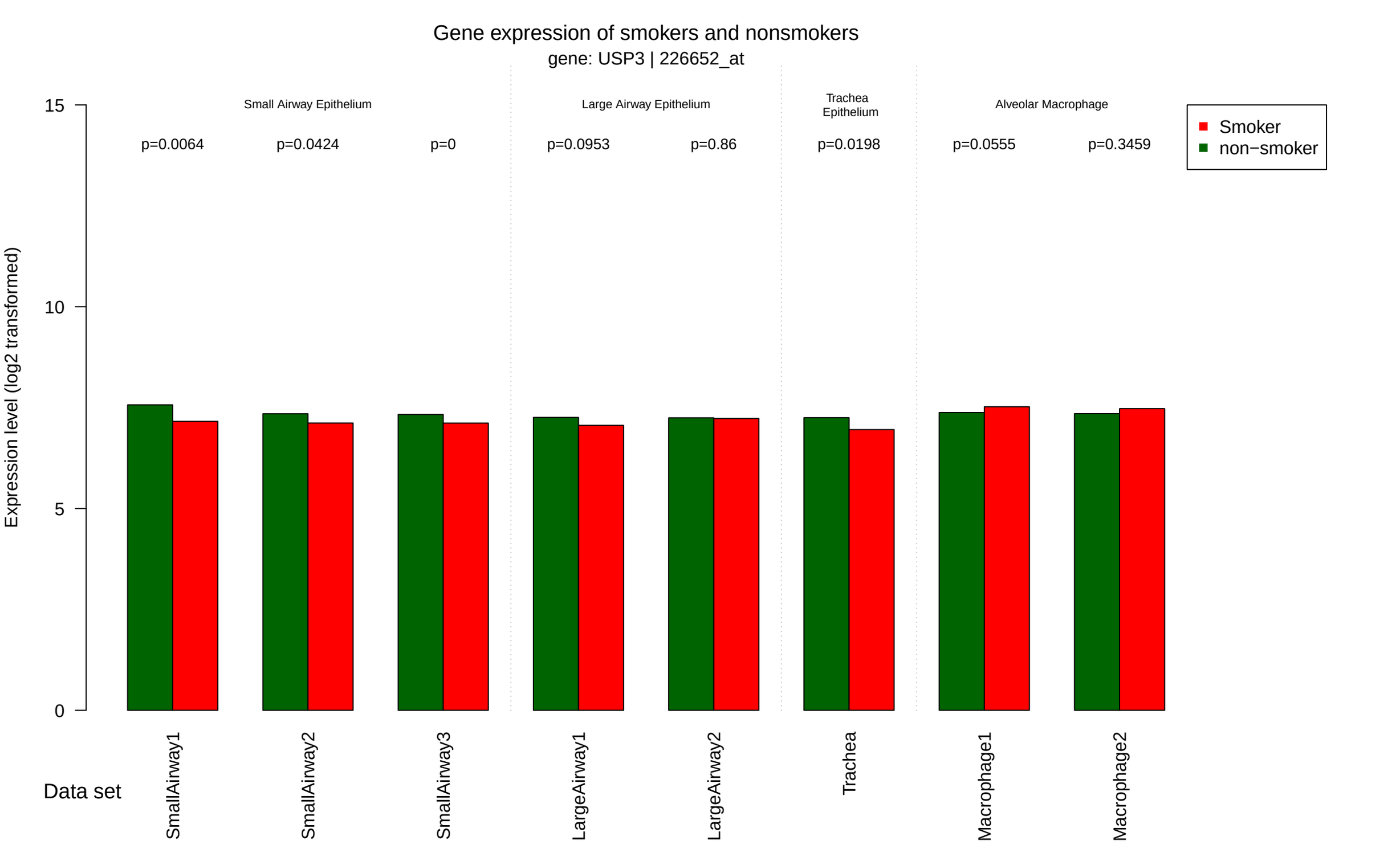
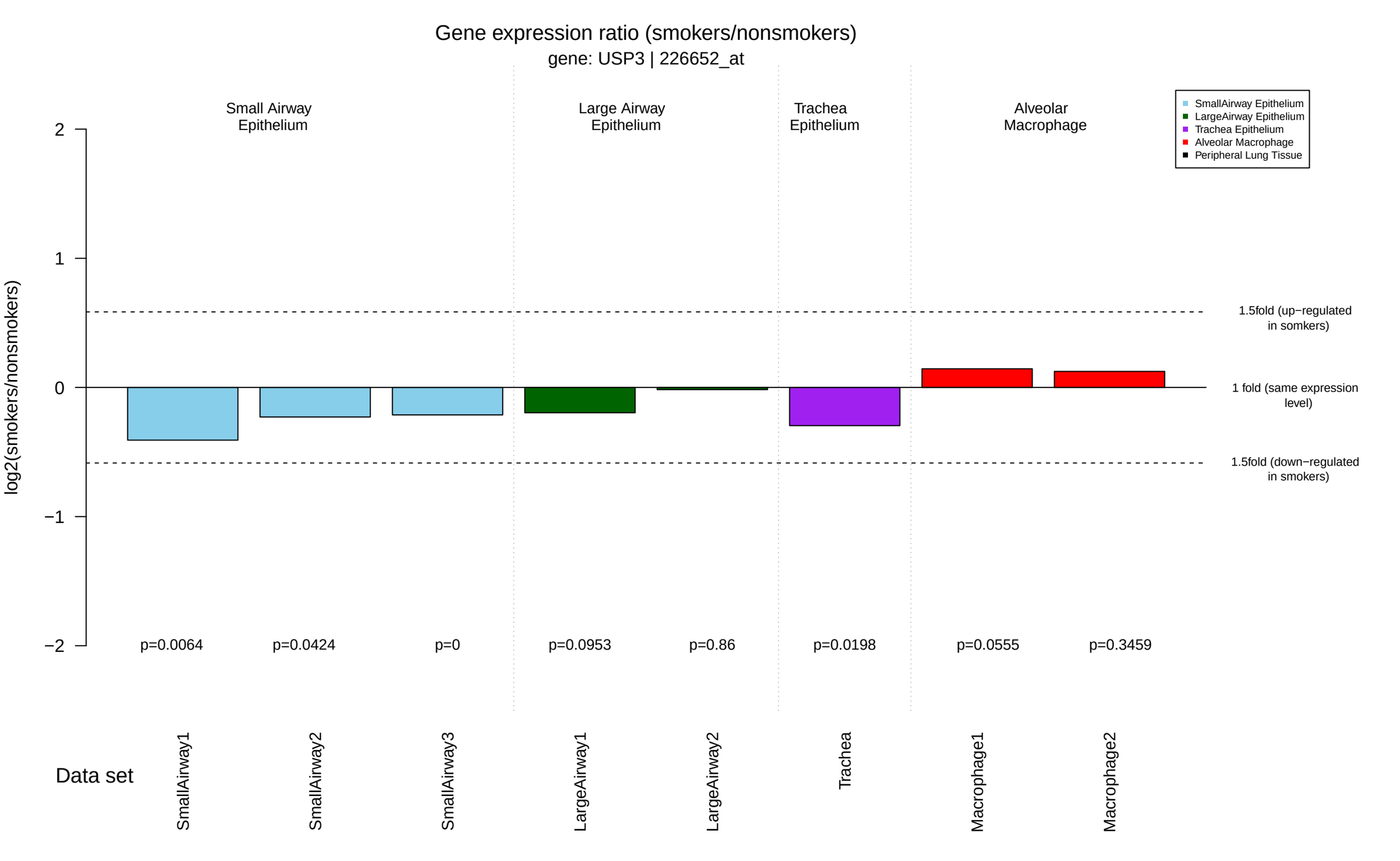
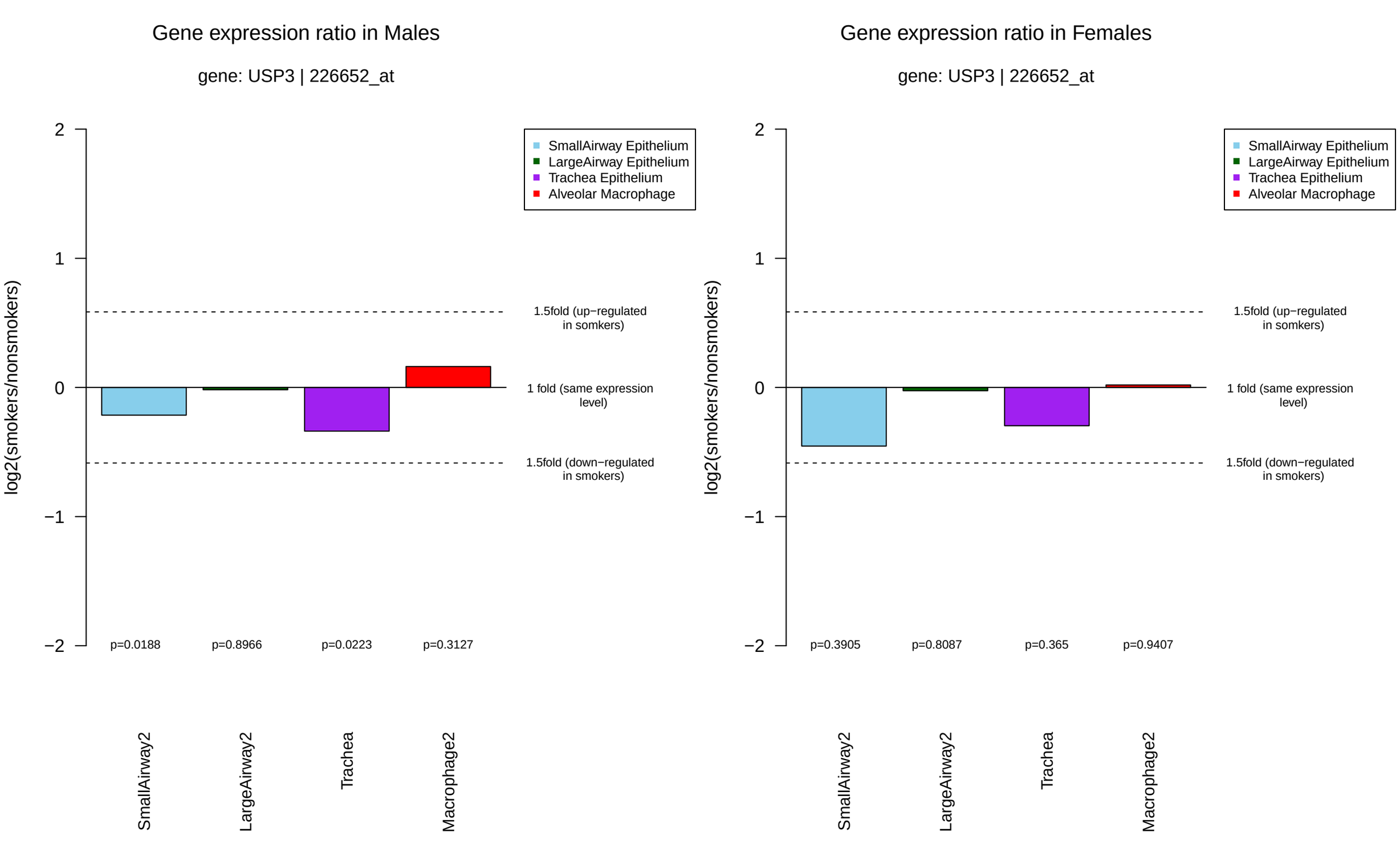
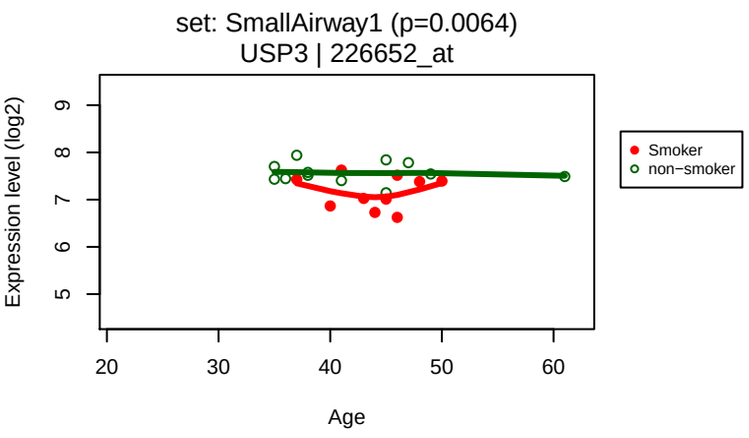
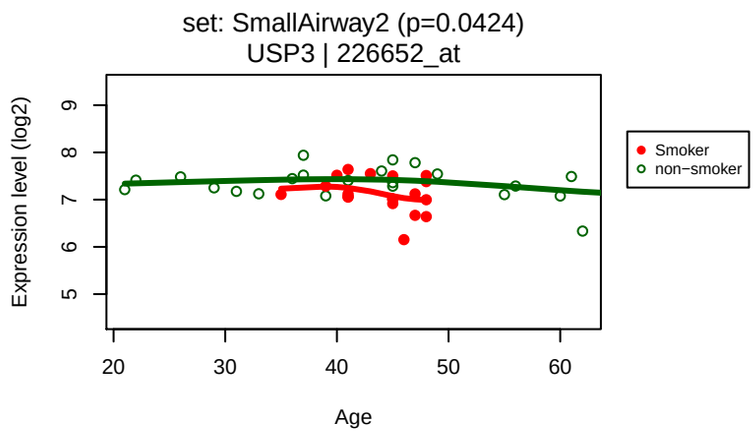
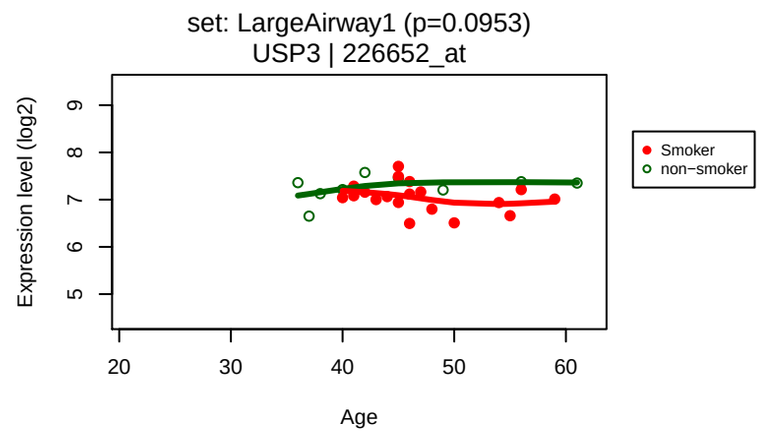

Gene expression of smokers and nonsmokers
gene: USP3 | 226652_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0064
p=0.0424
p=0
p=0.0953
p=0.86
p=0.0198
p=0.0555
p=0.3459
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

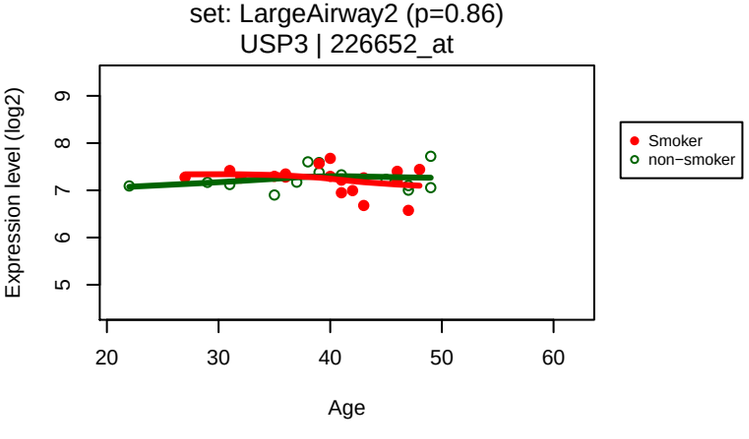

Gene expression ratio (smokers/nonsmokers)
gene: USP3 | 226652_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0064
p=0.0424
p=0
p=0.0953
p=0.86
p=0.0198
p=0.0555
p=0.3459
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

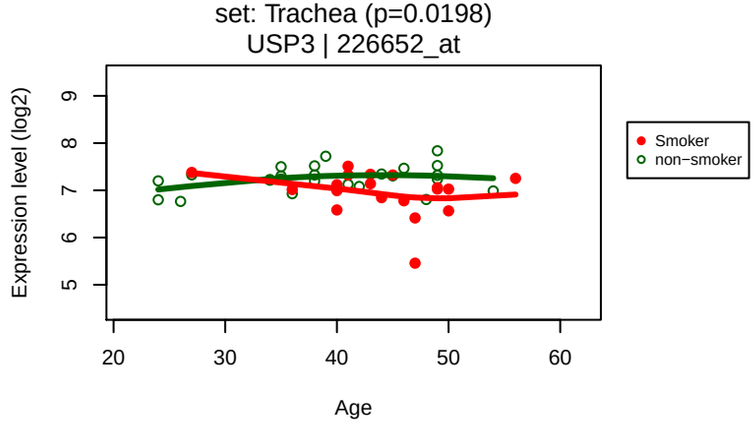

Gene expression ratio in Males
gene: USP3 | 226652_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0188
p=0.8966
p=0.0223
p=0.3127
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: USP3 | 226652_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3905
p=0.8087
p=0.365
p=0.9407
SmallAirway2
LargeAirway2
Trachea
Macrophage2

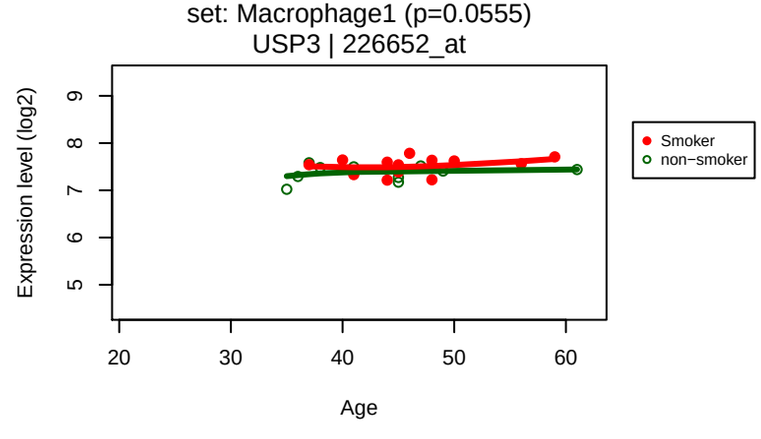

set: SmallAirway1 (p=0.0064)
USP3 | 226652_at
Expression level (log2)
Age
Smoker
non−smoker

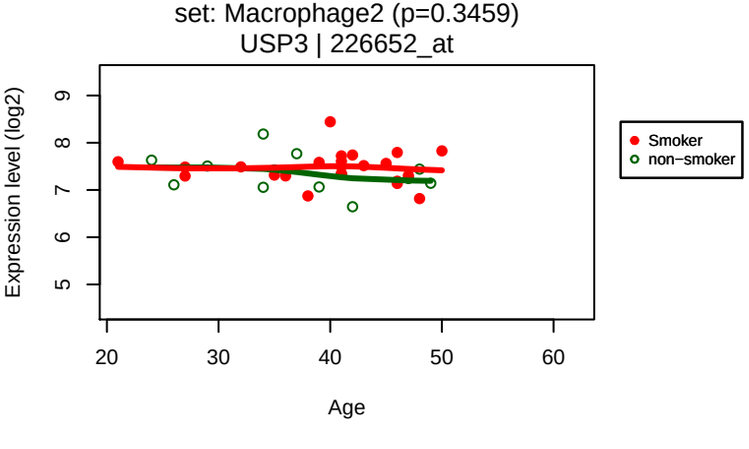

set: SmallAirway2 (p=0.0424)
USP3 | 226652_at
Expression level (log2)
Age
Smoker
non−smoker

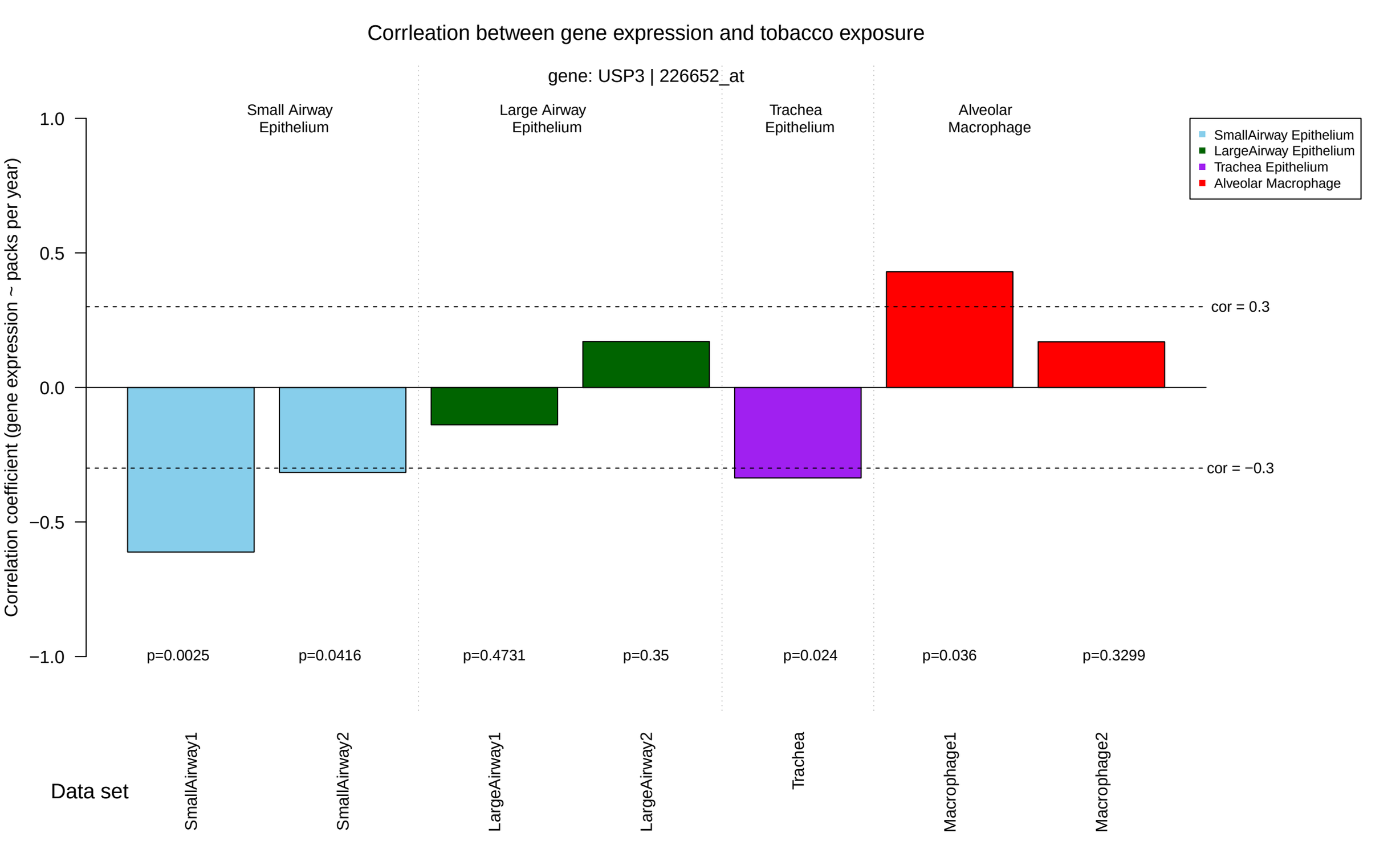

set: LargeAirway1 (p=0.0953)
USP3 | 226652_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.86)
USP3 | 226652_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0198)
USP3 | 226652_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0555)
USP3 | 226652_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3459)
USP3 | 226652_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: USP3 | 226652_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0025
p=0.0416
p=0.4731
p=0.35
p=0.024
p=0.036
p=0.3299
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2